

WONSAN, Korea, Democratic People's Republic of

WMO: 470550

Lat: 39.183N

Lon: 127.433E

Elev: 118

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	16.9	-13.9	2.6	19.7	-8.8	3.4	21.9	20.8	19.4	18.5	21.9	7.1	230	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.3	89.9	74.8	86.9	73.9	84.0	72.9	78.5	85.4	77.1	83.3	75.9	81.5	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.6	139.2	81.9	75.4	133.5	80.4	74.1	127.8	79.6	42.2	85.9	40.8	83.7	39.6	81.5	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	15.0	12.6	8.0	94.1	4.6	2.5	4.7	95.9	2.0	97.4	-0.6	98.8	-4.0	100.7
			5.4	81.2	3.9	1.8	2.6	82.5	0.3	83.5	-2.0	84.5	-4.8	85.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.4	28.9	32.6	41.6	52.9	62.0	68.4	74.6	75.3	67.3	57.8	44.7	33.4
	DBStd	17.26	6.42	6.15	7.17	7.46	7.12	5.76	6.07	4.71	4.57	5.48	7.45	6.87
	HDD50	2187	654	488	277	52	2	0	0	0	0	8	190	516
	HDD65	5137	1119	908	724	367	145	27	2	1	27	228	608	981
	CDD50	3433	0	0	18	140	375	551	763	784	519	251	32	0
	CDD65	908	0	0	0	6	53	129	300	319	96	5	0	0
	CDH74	6024	0	0	0	49	428	809	2275	2127	325	11	0	0
	CDH80	1743	0	0	0	5	97	251	787	556	47	0	0	0
Wind	WSAvg	4.4	5.0	4.9	5.3	5.0	4.4	3.8	3.7	3.6	3.6	4.0	4.8	5.1
Precipitation	PrecAvg	54.1	1.2	1.3	1.5	2.6	3.9	5.7	12.7	10.9	7.4	2.9	2.8	1.5
	PrecMax	80.6	4.3	5.2	3.9	6.3	10.6	14.2	33.9	25.4	17.2	15.2	23.8	7.4
	PrecMin	27.9	0.0	0.1	0.2	0.2	0.6	1.5	1.0	1.7	0.8	0.3	0.3	0.0
	PrecStd	9.7	1.0	1.1	1.1	1.6	2.6	3.9	6.3	5.2	4.3	3.1	4.5	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.4	52.7	67.8	79.3	87.5	91.4	93.5	92.8	84.8	75.3	67.0	53.0
		MCWB	38.2	41.9	52.2	58.2	64.0	68.4	76.6	78.0	71.2	60.9	53.7	43.6
	2%	DB	43.6	48.2	61.7	74.3	82.5	86.4	90.3	88.9	80.8	71.6	62.5	49.0
		MCWB	35.4	37.9	47.0	55.1	61.9	68.0	76.3	77.3	69.4	59.9	51.2	40.8
	5%	DB	40.7	44.9	57.4	70.2	78.9	82.4	87.7	85.5	77.4	69.1	58.8	45.8
		MCWB	33.5	36.6	44.2	53.0	60.2	68.0	75.5	76.1	68.5	59.4	48.9	38.7
	10%	DB	38.1	41.9	53.4	66.2	75.1	78.4	84.6	82.7	75.1	66.7	55.7	43.1
		MCWB	31.7	34.7	42.4	51.3	58.9	66.8	74.7	75.4	68.2	58.4	47.8	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.9	43.3	53.5	61.7	68.2	74.1	80.4	80.6	75.1	66.0	56.7	45.7
		MCDB	45.4	50.7	66.3	72.9	79.9	83.2	88.5	87.4	80.9	70.2	63.2	51.5
	2%	WB	36.4	39.6	48.9	57.8	64.7	71.8	78.3	78.9	72.6	63.6	53.6	42.1
		MCDB	41.7	46.0	59.1	69.9	78.1	79.6	86.1	85.6	77.2	68.6	59.7	47.4
	5%	WB	34.2	37.3	45.5	55.3	62.7	70.2	77.1	77.6	70.8	61.8	51.3	39.5
		MCDB	39.1	43.6	54.9	66.2	74.0	77.4	84.2	83.4	75.1	66.7	56.4	44.9
	10%	WB	32.5	35.3	43.0	52.6	61.0	68.8	75.8	76.5	69.2	59.9	49.0	37.2
		MCDB	36.9	40.8	51.7	63.4	70.4	75.3	82.6	81.5	73.4	64.7	54.6	42.0
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.1	13.1	15.1	14.8	11.3	9.3	9.3	11.5	13.3	12.4	11.9
		MCDBR	14.0	15.6	19.4	23.1	23.4	20.2	15.7	14.4	15.6	16.5	16.6	13.8
		MCWBR	10.3	10.0	10.5	11.2	10.2	8.9	6.8	6.8	8.8	9.9	10.8	10.3
	5% WB	MCDBR	12.3	14.0	17.5	20.6	19.2	15.1	12.7	11.4	11.9	13.2	13.7	12.8
		MCWBR	9.8	9.8	10.9	11.6	10.3	8.0	6.6	6.4	8.1	9.5	10.5	10.3
Clear-Sky Solar Irradiance	taub	0.393	0.458	0.532	0.601	0.613	0.671	0.631	0.547	0.485	0.492	0.452	0.394	
	taud	2.014	1.881	1.767	1.676	1.702	1.650	1.794	2.002	2.091	1.945	1.972	2.048	
	Ebn at Noon	231	231	228	222	223	210	218	234	240	219	208	220	
	Edh at Noon	40	52	65	75	75	79	68	54	46	49	42	36	
All-Sky Solar Radiation	RadAvg	775	1049	1354	1573	1720	1587	1296	1312	1259	1075	745	683	
	RadStd	63	84	132	144	179	159	192	147	141	89	102	69	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+122

Nomenclature: See separate page